

CS 506 lab 12 questions

May 2026

Questions

1. Consider a binary classification task requiring 2 input neurons, 4 total hidden neurons, and 1 output neuron. There will be fewer total parameters (i.e., weights and biases) if all the hidden neurons are placed into one hidden layer as opposed to two hidden layers
 - (a) True
 - (b) False
2. Consider a binary classification task requiring 1 input neuron, 5 total hidden neurons, and 1 output neuron. There are fewer total parameters in a network where all hidden neurons are in a single hidden layer compared to a network where the neurons are split into two hidden layers, where layer 1 has 2 neurons and layer 2 has 3 neurons, respectively.
 - (a) True
 - (b) False
3. The activation function $A(x) = x$ applies a non-linear transformation to the hidden layers
 - (a) True
 - (b) False
4. Even with a complex architecture, a neural network with no activation function in hidden layers is the same as a neural network with no hidden layers.
 - (a) True
 - (b) False
5. In deep networks with tanh or sigmoid activations, early layers struggle to learn because gradients become too small by the time they reach the first layer
 - (a) True

- (b) False
- 6. You train a fully connected network with tanh activations and sigmoid outputs. Input features have large positive means. What will happen?
 - (a) Most hidden features will be constant (either all 1 or all -1) on initialization and take a very long time to change because gradients are very small.
 - (b) It will work fine - neural networks can adapt to any features
 - (c) The network will crash because finding the derivative of the sigmoid function is not possible
- 7. Consider a single ReLU neuron with weights initialized from a symmetric distribution and data centered at 0. If the bias b is initialized to a small positive value (e.g. 0.01), then more than 50% of data points will make the neuron activate.
 - (a) True
 - (b) False

Solutions

1. False
2. True
3. False
4. True
5. True
6. (a)
7. True